

Eigenvectors

Definition: Eigenvectors

$A = [a_{ij}]$ be a square matrix of order 'n'

If there exist a non zero vector $X = \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{bmatrix}$

such that $AX = \lambda X$, then the vector X is called an Eigenvector of A corresponding to the Eigenvalue λ .

Problem: Find all the Eigenvalues and Eigenvectors of the matrix,

$$M = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

Solution:

Given that,

$$M = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} \dots \dots \dots (1)$$

1. To find the eigenvalues of M ;

$$M - \lambda I_3 = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{bmatrix} \dots \dots \dots (2)$$

For characteristic equation;

$$|M - \lambda I_3| = 0$$

$$|M - \lambda I_3| = 0 \Rightarrow \begin{vmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{vmatrix} = 0 \Rightarrow -\lambda^3 + 4\lambda^2 + \lambda - 4 = 0$$

Therefore the eigenvalues are, (by solving the characteristic equation)

$$\therefore -\lambda^3 + 4\lambda^2 + \lambda - 4 = 0 \quad \lambda = 4, -1, 1$$

2. To find the eigenvectors of M;

For eigenvectors $(M - \lambda I_3)X = 0 \dots \dots \dots (3)$

$$\text{Where, } X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ and } M - \lambda I_3 = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{bmatrix}$$

Therefore,

$$\begin{bmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \dots \dots \dots (4)$$

Now for, $\lambda = 1$ from equation (4), we get

$$\begin{bmatrix} 4-1 & 6 & 6 \\ 1 & 3-1 & 2 \\ -1 & -4 & -3-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ -1 & -4 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Back to equation, we get

$$3x_1 + 6x_2 + 6x_3 = 0$$

$$x_1 + 2x_2 + 2x_3 = 0$$

$$-x_1 - 4x_2 - 4x_3 = 0$$

$$x_1 + 2x_2 + 2x_3 = 0$$

$$x_1 + 2x_2 + 2x_3 = 0$$

$$-x_1 - 4x_2 - 4x_3 = 0$$

$$x_1 + 2x_2 + 2x_3 = 0$$

$$x_1 + 4x_2 + 4x_3 = 0$$

Solving by cross multiplication rule we get

$$\frac{x_1}{2 \times 4 - 2 \times 4} = \frac{x_2}{2 \times 1 - 4 \times 1} = \frac{x_3}{1 \times 4 - 2 \times 1}$$

$$\Rightarrow \frac{x_1}{0} = \frac{x_2}{-2} = \frac{x_3}{2}$$

Therefore, $X_1 = \begin{bmatrix} 0 \\ -2 \\ 2 \end{bmatrix}$

Now for, $\lambda = -1$ from equation (4), we get

$$\begin{bmatrix} 4+1 & 6 & 6 \\ 1 & 3+1 & 2 \\ -1 & -4 & -3+1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 5 & 6 & 6 \\ 1 & 4 & 2 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Back to equation, we get

$$5x_1 + 6x_2 + 6x_3 = 0$$

$$x_1 + 4x_2 + 2x_3 = 0$$

$$-x_1 - 4x_2 - 2x_3 = 0$$

$$5x_1 + 6x_2 + 6x_3 = 0$$

$$x_1 + 4x_2 + 2x_3 = 0$$

$$x_1 + 4x_2 + 2x_3 = 0$$

$$5x_1 + 6x_2 + 6x_3 = 0$$

$$x_1 + 4x_2 + 2x_3 = 0$$

Solving by cross multiplication rule we get

$$\frac{x_1}{6 \times 2 - 6 \times 4} = \frac{x_2}{6 \times 1 - 5 \times 2} = \frac{x_3}{5 \times 4 - 6 \times 1}$$

$$\Rightarrow \frac{x_1}{-12} = \frac{x_2}{-4} = \frac{x_3}{14}$$

$$\Rightarrow \frac{x_1}{-6} = \frac{x_2}{-2} = \frac{x_3}{7}$$

Therefore, $X_2 = \begin{bmatrix} -6 \\ -2 \\ 7 \end{bmatrix}$

Now for, $\lambda = 4$ from equation (4), we get

$$\begin{bmatrix} 4-4 & 6 & 6 \\ 1 & 3-4 & 2 \\ -1 & -4 & -3-4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 0 & 6 & 6 \\ 1 & -1 & 2 \\ -1 & -4 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Back to equation, we get

$$6x_2 + 6x_3 = 0$$

$$x_1 - x_2 + 2x_3 = 0$$

$$-x_1 - 4x_2 - 5x_3 = 0$$

$$x_2 + x_3 = 0$$

$$x_1 - x_2 + 2x_3 = 0$$

$$x_1 + 4x_2 + 5x_3 = 0$$

By solving, we get

$$\frac{x_1}{(-1) \times 5 - 2 \times 4} = \frac{x_2}{2 \times 1 - 5 \times 1} = \frac{x_3}{1 \times 4 - 1 \times (-1)}$$

$$\Rightarrow \frac{x_1}{-13} = \frac{x_2}{-3} = \frac{x_3}{5}$$

Therefore, $X_3 = \begin{bmatrix} -13 \\ -3 \\ 5 \end{bmatrix}$

The Eigenvalues are 1, -1, 4 and Eigenvectors are $\begin{bmatrix} 0 \\ -2 \\ 2 \end{bmatrix}$, $\begin{bmatrix} -6 \\ -2 \\ 7 \end{bmatrix}$, $\begin{bmatrix} -13 \\ -3 \\ 5 \end{bmatrix}$.

Assignment

1. Find all the Eigenvalues and Eigenvectors of the following matrices,

i. $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$

ii. $B = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

iii. $C = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

iv. $D = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$